



ECON3203: Econometric Theory and Machine Learning

Kindergarten's Reading and Math Scoring Models

Group PGI

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Summary of Task and Main Findings

The aim of this assignment was to analyse the factors that influence kindergarten reading and mathematics achievement and to develop accurate predictive models. The dataset included student performance scores along with a wide range of demographic, classroom, school, and teacher characteristics. The task involved four components: (1) conducting exploratory data analysis (EDA) to identify key patterns, (2) building predictive models for reading and math scores, (3) selecting the best-performing model using cross-validation, and (4) interpreting the main drivers of performance using SHAP values and economic reasoning.

The EDA revealed several strong patterns. Students eligible for free lunch (a proxy for low socio-economic status) consistently scored lower in both reading and math. Smaller class types were associated with higher achievement, which is consistent with findings from the education literature. Teacher characteristics also played a role: greater experience and higher ladder level generally related to better performance, although there was still wide variation between students. Ethnicity differences were observed as well, particularly lower scores among Native American students, highlighting ongoing disadvantages. Overall, the distributions of reading and math scores were close to normal, suggesting that several modelling approaches were appropriate.

A wide range of models were tested, including OLS, ridge, lasso, elastic net, random forest, gradient boosting, and XGBoost. Standard linear models, even with interactions, performed poorly due to low explanatory power (R^2 around 0.14 - 0.17) and multicollinearity. Regularised models gave slight improvements but were still limited. Tree-based models performed best because they could capture nonlinear relationships. For reading, the tuned XGBoost model achieved the lowest MSE (≈ 778.9) and the highest R^2 (≈ 0.212). For math, the tuned XGBoost model again performed best, with MSE ≈ 1773.2 and $R^2 \approx 0.22$.

SHAP analysis provided insight into why the XGBoost model made its predictions. In both subjects, socio-economic status (free lunch) was the most influential factor. Teacher experience, class size, and gender were also impactful. Male students tended to score slightly lower, and smaller classes had clear benefits. Teacher degree level and teacher ethnicity (especially Caucasian teachers) showed additional influence, and rural schools were generally associated with lower performance. These findings align with trends commonly reported in the education economics literature.

Overall, the results show that linear models struggle because education outcomes involve nonlinear relationships. Machine learning methods - especially tuned XGBoost - produced noticeably better predictions. SHAP proved helpful in explaining the classroom, demographic, and socioeconomic factors that shape student performance.

1. Exploratory Data Analysis

1.1 Overview of the Dataset

In order to get started on modelling, it is first important to understand the relationship between each feature and the target variables. Before correlation can be found, we must also treat the data for any missing values. Table 1.1.1 was generated to provide an initial descriptive overview of the dataset.

	score_read	score_math	gender	birth	lunch	ethnicity	class_type	school	degree	ladder	experience	t_ethnicity	schooldistrict_id	school_id
count	4643.000000	4706.000000	5060	5055	5039	5059	5060	5060	5041	4549	5041.000000	5008	5060.000000	5060.000000
unique	NaN	NaN	2	15	2	6	3	4	4	5	NaN	2	NaN	NaN
top	NaN	NaN	male	1980 Q3	non-free	cauc	regular+aide	rural	bachelor	level1	NaN	cauc	NaN	NaN
freq	NaN	NaN	2591	1333	2578	3389	1786	2351	3306	3737	NaN	4193	NaN	NaN
mean	436.272238	485.123459	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	9.277723	NaN	18.699209	40.042885
std	31.358938	47.674836	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	5.854946	NaN	11.046311	22.770969
min	346.000000	288.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0.000000	NaN	1.000000	1.000000
25%	414.000000	454.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	5.000000	NaN	11.000000	21.000000
50%	433.000000	478.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	9.000000	NaN	15.000000	39.000000
75%	453.000000	513.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	13.000000	NaN	27.000000	60.000000
max	627.000000	626.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	27.000000	NaN	42.000000	80.000000

Table 1.1.1: Summary of the data.

The descriptive summary table provided an overview of the dataset, including the count of observations, the number of unique values, the most frequent entries, and the five-number summary (minimum, first quartile, median, third quartile, and maximum) for each variable. Since most features were categorical the five-number summary in the last 5 rows of this table was not informative. We found that there were missing values across the data, most importantly in the **score_read** and **score_math** columns.

To get an insight into the nature of the missing values in each column, table 1.1.2 was used. No feature has a missing rate of over 10%. This means dropping rows with missing values won't cause too much information loss. A key observation at this stage was that the patterns of missingness in **score_read** and **score_math** were not systematically related. Specifically, instances were identified in which valid values for **score_read** coexisted with missing entries for **score_math** (and vice-versa), indicating that the two variables exhibited independent mechanisms of missing data. To conclude, rows with missing values in columns **experience**, **lunch**, **t_ethnicity** were dropped completely. More importantly, two dataframes were created, **read** and **math**. The motivation for this was to deal with the independence of missing values in these two columns, by training two different models for **score_read** and **score_math** the amount of information the models were trained on was maximised. Birth, school_district

	Missing Count	Missing Rate (%)
score_read	417	8.24
score_math	354	7.00
gender	0	0.00
birth	5	0.10
lunch	21	0.42
ethnicity	1	0.02
class_type	0	0.00
school	0	0.00
degree	19	0.38
ladder	511	10.10
experience	19	0.38
t_ethnicity	52	1.03
schooldistrict_id	0	0.00
school_id	0	0.00

Table 1.1.2: Missing Values Count and Rate (%)

1.2 Visualisation and Initial Insights

1.2.1 Reading Scores

A histogram was produced to ensure that the reading scores were approximately normally distributed. The following plot shows that the distribution is nearly normal, centred at about a score of 425.

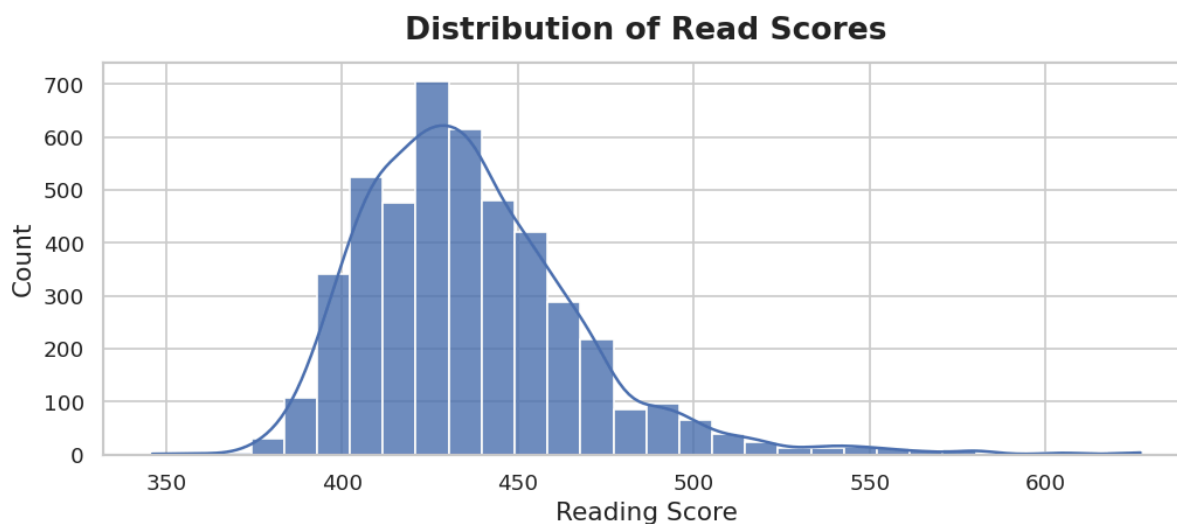


Figure 1.2.1: Distribution of Read Scores

1.2.2 Math Scores

Math scores follow a similar distribution as seen by the following plot. Normally distributed centred at a score of around 475. Knowing the distribution of the two scores being predicted is important to ensure the modelling approaches being used in the future will be useful.

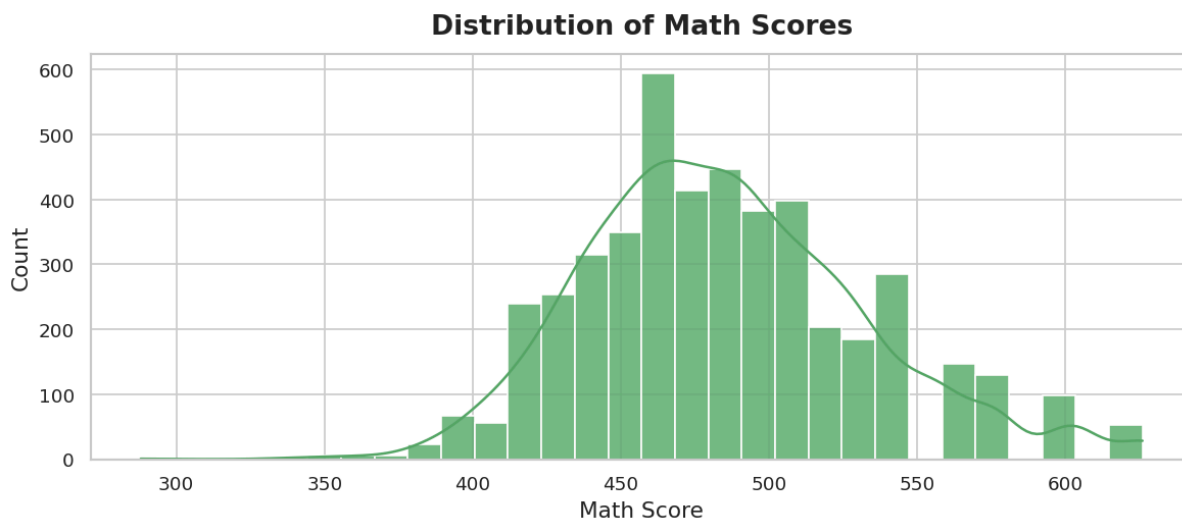


Figure 1.2.2: Distribution of Math Scores

1.2.3 Gender

It seems as though there is a mild correlation between a student's reading scores and their gender. It is clear to see that female students had higher minimum, Q1, Median and Q3 scores. However, the two are relatively closely distributed. In a similar way, student math scores are slightly elevated in female students however only the Q3 and maximum scores are significantly higher. May imply that the female scores are negatively skewed in the math scores.

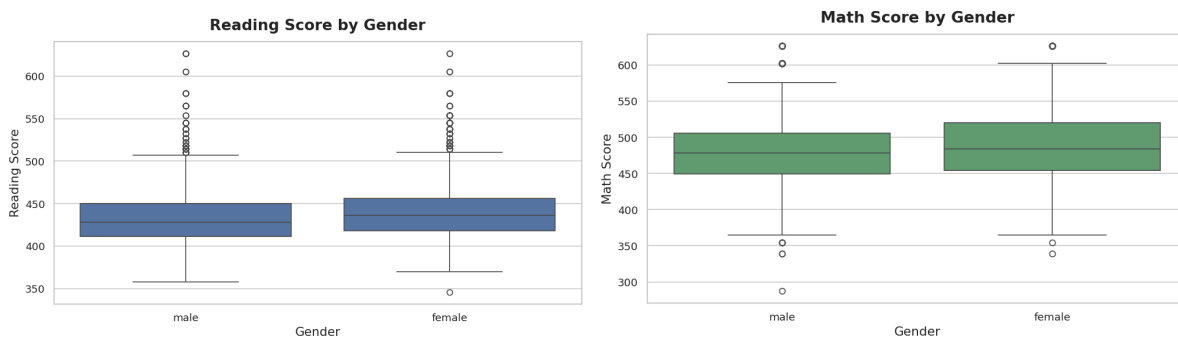


Figure 1.2.3: Reading Score and Math Score by Gender

1.2.4 Student Ethnicity

When analysing the plot of both math and reading scores classified by student ethnicity the most important feature is the significantly lower and closely grouped scores for the ethnicity **amindian** which represents students of Native-American descent. The simplest explanation for this is the well-documented socio-economic disadvantage native americans face, leading to a lack of resources, funding etc.

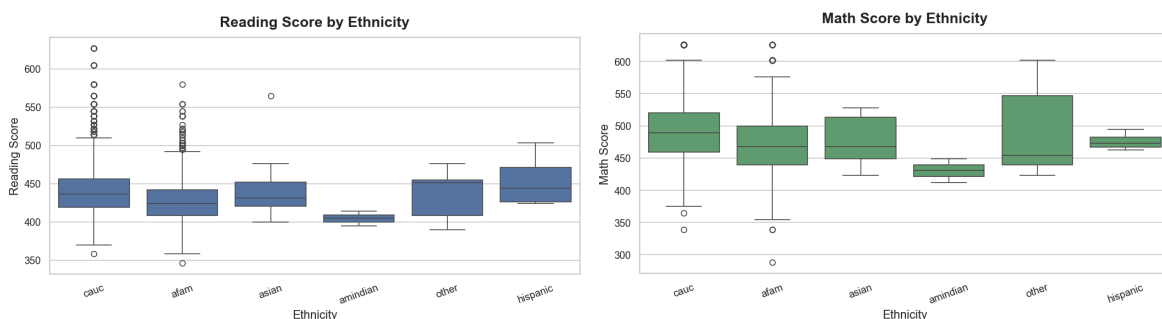


Figure 1.2.4: Reading Score and Math Score by Ethnicity

1.2.5 Free Lunches

The plots for the variable **lunch** revealed that schools with free lunches tend to have students that perform worse than those that do not. It is possible that free-lunches are a proxy for low socio-economic status. Students attending these schools often face reduced access to educational materials, less stable learning environments etc.

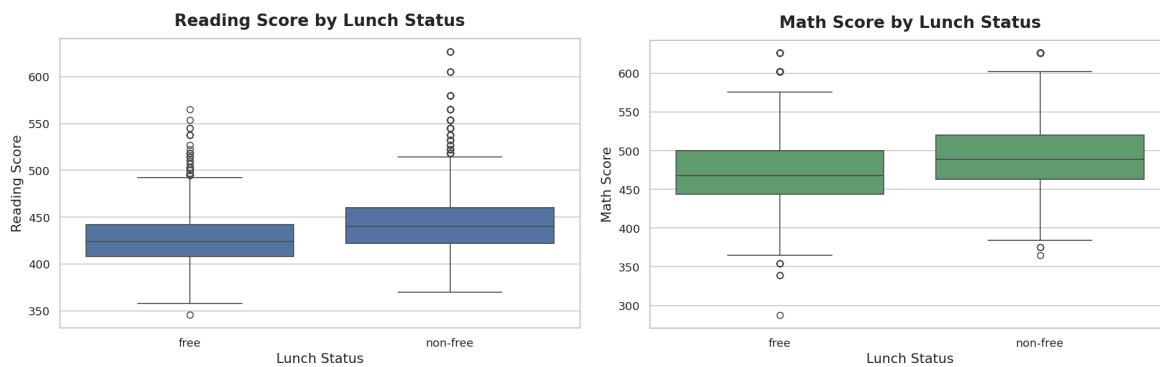


Figure 1.2.5: Reading Score and Math Score by Lunch Status

1.2.6 Class Type

It seems that students attending classes with smaller sizes perform better than the students that do not. The likely reason for this is the increased individual focus placed on each child. With fewer students competing for time and support, teachers can tailor instruction more effectively, leading to improved academic outcomes.

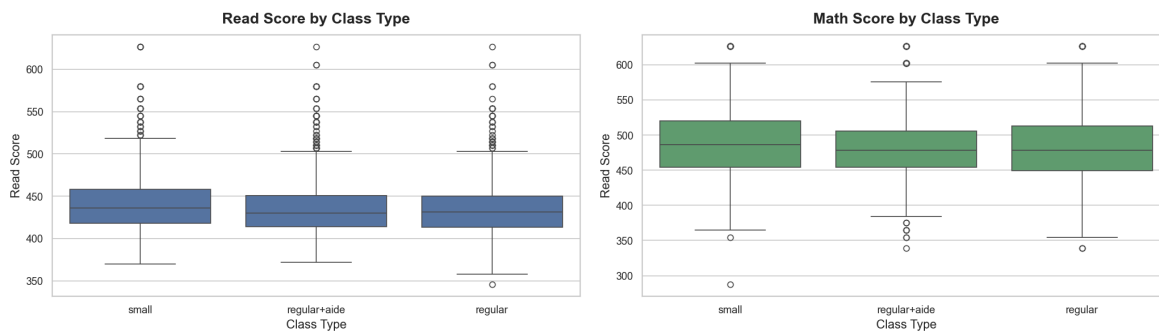


Figure 1.2.6: Reading Score and Math Score by Class Type

1.2.7 School Types

Box plots of read and math scores by school type revealed that inner-city schools have slightly worse performance overall. We were unable to pinpoint a robust hypothesis for why this may be the case.

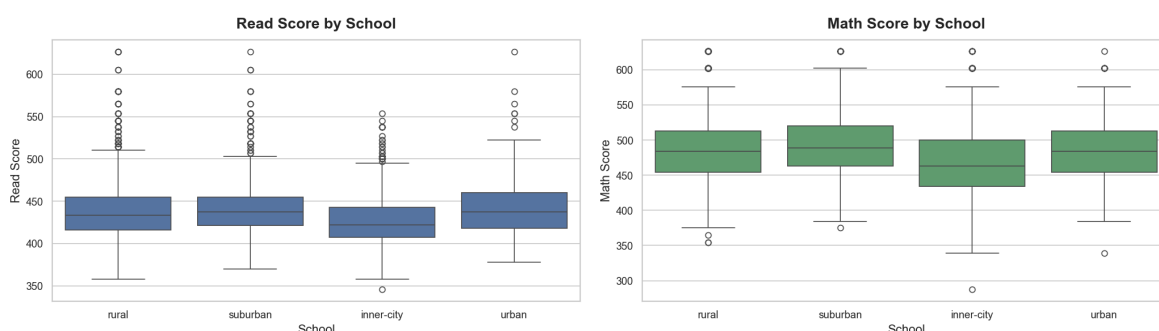


Figure 1.2.7: Reading Score and Math Score by School

1.2.8 Teacher Degree

Teachers with specialist degrees seem to have a stronger influence on students' scores in both fields. This may be because specialist teachers are more accessible to schools that are "richer,"

and it could also reflect a combination of higher teacher quality and a better overall student environment.

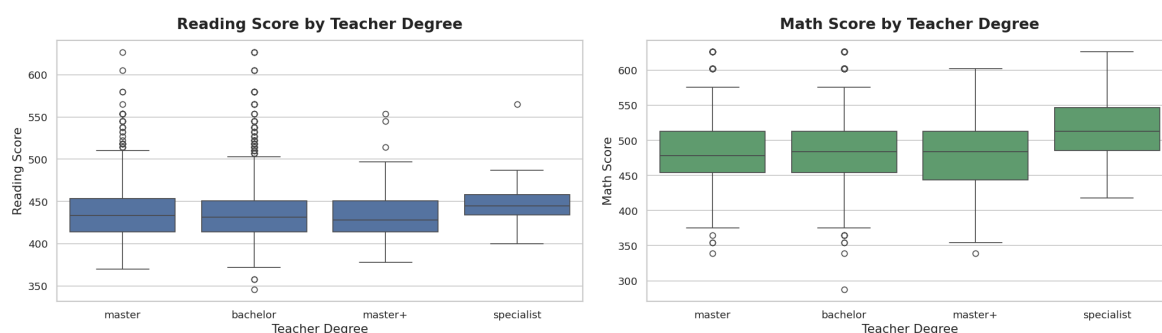


Figure 1.2.8: Reading Score and Math Score by Teacher Degree

1.2.9 Teacher Ladder

Teacher ladder, is akin to **experience** as they both increase with the amount of time a teacher spends teaching. As expected, going from probation up to level3 increases the overall score attained by students. More experienced teachers may have better teaching methods or philosophies and as a result students may learn better.

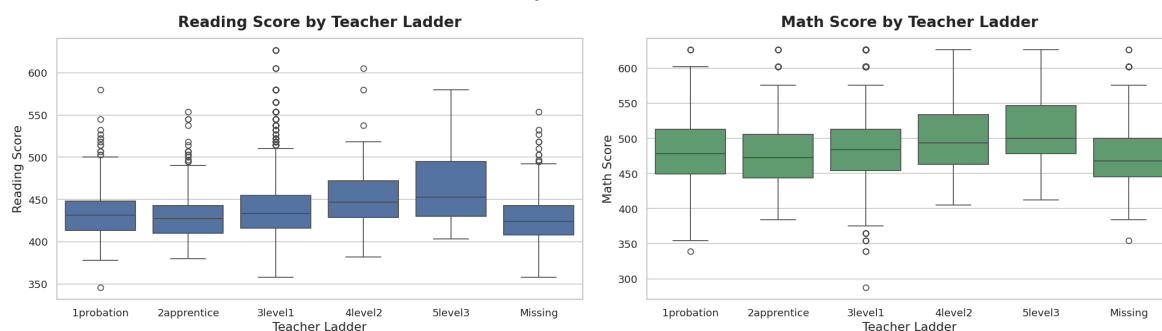


Figure 1.2.9: Reading Score and Math Score by Teacher Ladder

1.2.10 Teacher Experience

The scatter plots comparing teacher experience with student reading and math scores show a mild positive relationship. As teachers gain more years of experience, there is a slight upward trend in both subjects, suggesting that seasoned teachers may employ more effective instructional strategies or classroom management techniques. However, the relationship is not strongly linear, indicating that while experience contributes to better student outcomes, it is likely one of several factors influencing performance.

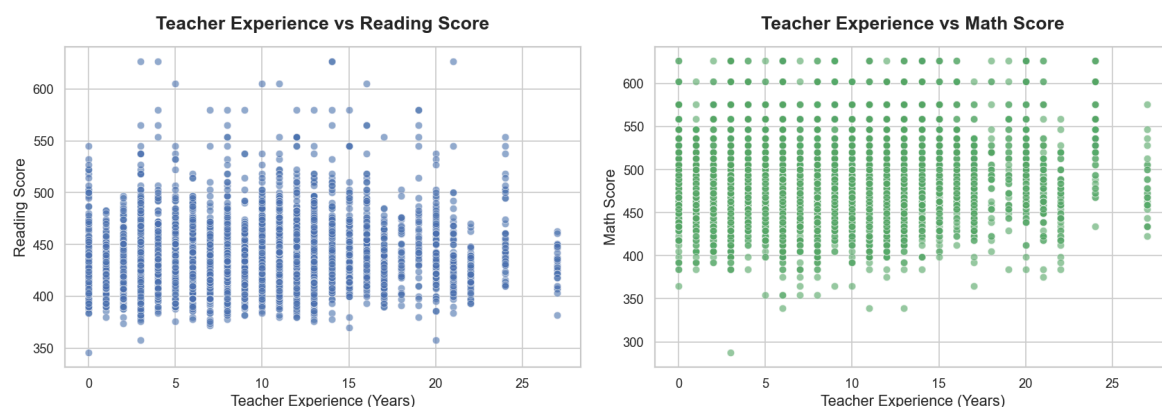


Figure 1.2.10: Reading Score and Math Score by Teacher Experience

1.3 Key Observations

- Teachers with more years of experience tend to have students with slightly higher reading and math scores.
- The relationship is positive but weak, indicating that experience alone does not fully explain student performance.
- There is high variability in scores at all experience levels, suggesting other factors (school type, resources, student background) play a major role.
- No sharp jumps or thresholds appear, improvements linked to experience seem gradual rather than dramatic.

2. Models (Reading Score)

2.1 Overview of Models Tested

To predict kindergarten reading achievement, we estimated a range of linear and non-linear models with increasing flexibility. The goal was to develop predictive models that balance accuracy with interpretability while addressing known challenges in the dataset such as many categorical variables, multicollinearity, and the presence of complex, potentially non-linear relationships.

Pre - processing

1. All categorical variables were one-hot encoded (dropping reference categories).
2. A broad set of interaction terms was generated among selected student, teacher, and school variables.
3. For regularised models, numerical variables and interaction terms were standardised.
4. Models were trained using a 70/30 train, test split and 5-fold cross-validation (CV) where applicable.

Models included

1. Ordinary Least Squares (OLS)
 1. Full model including all main effects and interaction terms.
 2. Serves as the baseline and offers interpretability.
2. Regularised Linear Models
 1. Ridge regression (L2 penalty)
 2. Lasso regression (L1 penalty)
 3. Elastic Net (combined L1/L2 penalty)

These models address multicollinearity and high dimensionality.
3. Tree-Based Models
 1. Random Forest
 2. Gradient Boosting (GBM)
 3. XGBoost

Tree models can naturally capture complex non-linearities and interactions without feature engineering. For each model, out-of-sample performance was evaluated using R squared and mean squared error (MSE), with 5-fold cross-validation applied to prevent overfitting test set evaluation.

2.2 Model Evaluation and Selection

*Except for OLS, test R^2 and MSE represent the average scores across 5-fold cross validation.

Model	Test R^2	Test MSE
OLS	0.144	873.27
Ridge	0.169	852.96
Lasso	0.167	855.22
Elastic Net	0.168	854.09

Table 2.2.1: Table of Models Comparison based on 5-fold cross-validation Test R-squared and Test MSE

Findings

1. OLS performs the worst due to severe multicollinearity and many noisy interaction terms.
2. All regularised models improve on OLS.
3. Ridge regression achieves the best performance among linear models, consistent with a non-sparse signal structure where many predictors have small but real effects.

Tree-Based Models

Model	Initial Test MSE	Mean CV R^2	Mean CV Test MSE
Random Forests	813.71	0.142	847.56
Gradient Boosting	804.96	0.204	786.62
Xgboost	801.13	0.212	778.87

Table 2.2.2: Table of Tree Models Comparison based on 5-fold cross-validation Test R-squared and Test MSE

Random Forests

1. Test $R^2 = 0.207$
2. Test $MSE = 813.71$
3. CV mean $MSE = 847.56$

The Random Forest performs much better than any of the linear models, but its results vary a lot across the CV folds. This suggests the model is quite sensitive to the large number of dummy variables and the uneven distribution of some categories.

Gradient Boosting

1. Test $MSE = 804.96$
2. CV mean $MSE = 786.62$
3. CV $R^2 = 0.204$

The tuned GBM model performs more consistently than Random Forest and achieves a strong reduction in *MSE* relative to all previous models.

XGBoost

1. Initial test *MSE* = 801.13
2. After tuning and CV:
 - a. Mean CV *MSE* = 778.87
 - b. Mean CV R^2 = 0.212

Among all models evaluated, XGBoost demonstrated the lowest mean squared error (MSE), establishing it as the most effective algorithm for predicting reading scores. To further enhance its performance, hyperparameter tuning was conducted to identify the optimal combination of parameters tailored to the training dataset. A parameter grid search was employed to systematically explore key hyperparameters, including:

- **Max_depth**: Controls the maximum depth of each tree
- **Learning_rate**: Determines the step size at each iteration
- **Min_child_weight**: Minimum sum of instance weight needed in a child
- **Subsample**: Fraction of observations to be randomly sampled for each tree
- **Colsample_bytree**: Fraction of features to be randomly sampled for each tree

This tuning process aimed to minimize the model's prediction error and ensure robust generalization across unseen data.

SHAP Interpretation (XGBoost)

SHAP values highlight the most influential predictors:

1. lunch_nonfree (SES proxy)
2. experience
3. gender_male
4. class_type_small
5. ethnicity_cauc
6. degree_master
7. school_rural / school_suburban

The direction of effects aligns with expectations and the STAR literature:

students in small classes and those not eligible for free lunch tend to score higher, male students score lower on average, and rural schools underperform.

2.4 Justification for Selected Model

We have chosen the XGBoost (tuned with 5-fold CV) model:

Mean CV *MSE* = 778.9: the best among all models tested.

Reasons for Selection

1. **Best predictive accuracy**
XGBoost reduces out-of-sample error by ~90 points relative to OLS and ~75 points relative to Ridge, the best linear model.
2. **Ability to model non-linearities**
Educational achievement likely depends on complex interactions across SES, school characteristics, and teacher experience. XGBoost learns these patterns without requiring manually created interaction terms.
3. **Robustness through cross-validation**
XGBoost improves stability across folds compared to Random Forest, indicating better generalisation.
4. **Interpretability using SHAP**
Despite being a machine-learning model, XGBoost lets us see transparent feature attribution. SHAP values clearly identify the most influential factors, allowing the model to be both predictive and interpretable.
5. **Alignment with assignment goals**
The task requires both predictive performance and the ability to discuss key drivers of achievement. XGBoost excels in both areas.

3. Models (Math Score)

3.1 Overview of Models Tested

To analyse mathematics achievement, we followed the same structured modelling used for reading scores. This maintains consistency and allows direct comparison across outcomes, while incorporating both interpretable regression techniques and more flexible machine learning models.

Pre - processing

1. Dropped Observations missing score_math
2. One-hot encoded all categorical variables (e.g., school type, ethnicity, degree class type)
3. Generated a comprehensive set of pairwise interaction terms among student, teacher, and classroom variables.
4. Standardised numerical predictors for all regularised linear models
5. Used a 70/30 train-test split for initial evaluation and 5-fold cross validation (CV) for final model tuning.

Models Estimated

We evaluated five broad model classes:

1. Ordinary Least Squares (OLS)
 - Full model with all main effects + interactions
 - Serves as an interpretable baseline
2. Regularised Linear Models
 - Regularised Linear Models

- Lasso
 - Ridge
 - Elastic Net
3. Tree-Based Methods (No interactions needed)
- Decision Tree Regression
 - Random Forest
 - Gradient Boosting(GB)
 - XGBoost

These models can learn non-linearities and interactions automatically.

For all models, we record out-of-sample performance via R^2 and MSE .

3.2 Model Evaluation and Selection

Ordinary Least Squares (OLS)

The OLS model included over 80 predictors and showed severe multicollinearity:

- Condition number $\approx 1.2 \times 10^{16}$
- Many inflated standard errors and unstable coefficients

Performance:

- Test $R^2 = 0.145$
- Test $MSE = 2105.26$

This baseline reveals that linear relationships alone explain little of the variation in math scores.

Regularised Linear Models

*Except for OLS, test R^2 and MSE represent the average scores across 5-fold cross validation.

Model	Test R^2	Test MSE
OLS	0.145	2105.26
Ridge	0.144	1887.19
Lasso	0.144	1886.75
Elastic Net	0.144	1886.75

Table 3.2.1: Table of Models Comparison based on 5-Fold cross-validation Test r-squared and Test MSE

Key Observations

- All regularised models drastically improve on OLS, reducing error by ~ 200 points.

- However, R^2 remains between 0.144 and 0.145, meaning linear structure is weak.
- Lasso and Elastic Net show near-identical performance, indicating the math predictors form a dense signal rather than a sparse one.

This suggests that non-linear models are necessary to capture math achievement patterns.

Decision Tree Regressor

GridSearch over max depth and min samples per leaf:

- Best CV $MSE \approx 1980$ – 2050 , R^2 around 0.12–0.15.
- Underperforms ensemble methods.

Single trees are too unstable and cannot generalise well.

Random Forest

Initial RF (1000 trees, tuned max_features and min_samples_leaf):

- Test $R^2 = 0.196$
- Test $MSE = 1771.69$

This is a major improvement over all linear models.

However, after applying 5-fold CV with tuned hyperparameters:

Fold	R^2	MSE
Best Fold	~0.17	~1868
Worst Fold	~0.135	~1988

Table 3.2.2: Table of Random Forest Model performance comparison between the best fold and the worst fold in cross-validation.

- CV Mean $R^2 = 0.1555$
- CV mean $MSE \approx 1923.23$

The drop from the single train-test split indicates that RF was overfitting, and even after tuning it still underperforms compared to the boosting models.

Gradient Boosting (GBM)

Initial model:

- Test $R^2 = 0.191$
- Test $MSE = 1783.43$

After tuning (max_depth, subsample, learning_rate):

CV results:

- Mean $R^2 = 0.2129$
- Mean $MSE = 1792.35$

GBM consistently outperforms both linear models and random forest, and shows much more stable fold-to-fold behaviour.

Gradient Boosting (GBM)

Initial model:

- Test $MSE \approx 1778$

After tuning via 5-fold CV:

Fold	R^2	MSE
Fold 1	0.2000	1728.91
Fold 2	0.2596	1739.84
Fold 3	0.2260	1738.28
Fold 4	0.2536	1722.60
Fold 5	0.2021	1870.88

Table 3.2.3: Table of performance from 5-fold Cross-validation of Gradient Boosting

- Mean CV $R^2 = 0.2213$
- Mean CV $MSE = 1773.23$

This is the best performance of any model, and also the lowest variation across folds.

SHAP Analysis (Interpretability)

Using SHAP values from the final XGBoost model, the strongest predictors of math achievement are:

1. lunch_nonfree (SES)
2. experience (teacher experience)
3. gender_male (negative impact)
4. class_type_small
5. ethnicity_cauc
6. school_rural (negative)
7. degree_master
8. teacher ethnicity (t_ethnicity_cauc)

Key insights:

- SES (free lunch) is the strongest and most consistent driver, echoing education-economics literature.
- Small class size again shows positive influence, consistent with Project STAR findings.
- Male students tend to score lower, similar to reading.
- Teacher experience improves achievement across both outcomes.
- Rural schools underperform, reflecting structural inequalities.

3.3 Justification for Selected Model

Selected Model: Tuned XGBoost (5-Fold CV)

- Mean CV $MSE = 1773.23$ (lowest of all models)
- Mean CV $R^2 = 0.2213$ (highest of all models)

Why XGBoost is Optimal

1. Highest predictive accuracy

No other model, linear or non-linear, approaches the performance of tuned XGBoost.

2. Captures complex non-linear relationships

Math achievement is strongly influenced by nonlinear patterns and interactions (SES × class type, experience × school setting, etc.).

XGBoost learns these automatically.

3. Generalisation

Unlike Random Forest (which overfit under train-test but collapsed under CV), XGBoost remains stable across folds.

4. Interpretability using SHAP

The model offers clear explanations of feature contribution, supporting high-quality discussion of key drivers.

5. Aligned with assignment criteria

The task prioritises:

- predictive accuracy
- transparent interpretation
- thorough model comparison

XGBoost excels on all three dimensions.

Conclusion

Tuned XGBoost is selected as the final model for predicting math scores, achieving the lowest cross-validated $MSE(1773.23)$ and best overall predictive performance. It also enables robust interpretability through SHAP, highlighting SES, teacher experience, gender, class size, and school context as the predominant determinants of mathematics achievement.

4. Drivers of Performance

4.1 Feature Importance Analysis

The SHAP summary plot reveals which features have the strongest influence on the XGBoost model's predictions and how they affect the predicted reading score. A few clear patterns emerge. Note that the plot provided here is the SHAP summary plot from the read prediction model, the math prediction model summary is available in the notebook.

1. Lunch status (lunch_non-free) is the strongest predictor

- It appears at the top, indicating the largest average SHAP magnitude.

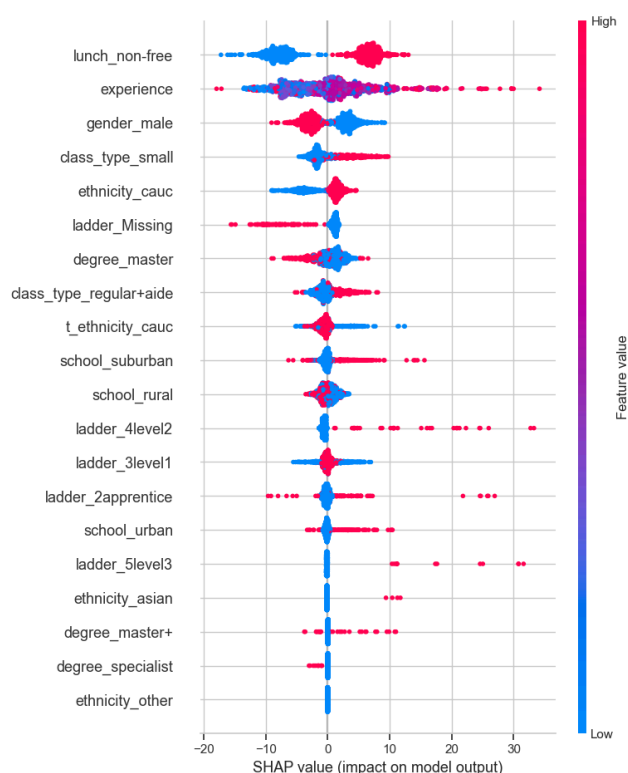


Figure 4.1.1 SHAP Summary plot on Feature Importance for score_read model

- Students with non-free lunch (high feature value = red) generally push predictions upward, increasing expected reading scores.
- Students receiving free lunch (blue) contribute negatively, consistent with socio-economic disadvantage.

2. Teacher experience is highly influential

- Higher experience values (red) tend to **increase predicted scores**.
- Lower experience (blue) is associated with negative SHAP values.

- This aligns with earlier EDA showing a mild positive correlation.

3. Gender (gender_male) shows a noticeable effect

- Being male (high feature value) generally lowers predicted scores, as seen by the cluster of red points on the negative side.
- Female students (blue) mildly increase predicted values.

4. Class size matters (class_type_small)

- Small classes (high/True) mostly increase reading predictions, suggesting students benefit from more individual attention.
- Regular-aide classes have a weaker and more mixed effect.

5. Ethnicity effects are present but moderate

- Caucasian ethnicity (ethnicity_cauc) shows a slight positive influence.
- Other ethnicities (Asian, Other, etc.) appear lower on the list, meaning they have weaker average influence on predictions.
- The model does not assign strong predictive power to these categories relative to socioeconomic or class-based features.

6. Teacher ladder and degree levels have small but consistent effects

- Higher ladder levels (e.g., level2, level3) generally push predictions upward.
- Missing ladder values have mixed effects, usually slightly negative.
- Teacher education (master's, master+, specialist) influences the model but far less than experience or lunch status.

7. School type (suburban, rural, urban) has modest influence

- Suburban and urban schools tend to contribute positively.
- Rural schools show mixed impacts but generally smaller absolute SHAP values.

4.2 Omitted Variables

The only variables not fed to any of the models were birth, schooldistrict_id, school_id. We did not believe that birth year or birth quarter had an effect on student performance. And our exploratory plots seemed to re-inforce this idea.

Similarly, schooldistrict_id and school_id were excluded because they behave as identifiers rather than information-rich explanatory features. These variables do not contain interpretable information about students or teachers; instead, they simply encode arbitrary labels for districts and schools. Including them would risk introducing noise, causing the model to overfit to specific IDs, reducing generalisability to unseen schools, and inflating the feature space without adding predictive value.

Since these identifiers have no intrinsic numerical or ordinal meaning, and their categories do not reliably map to any underlying socioeconomic or educational attributes, they were dropped from all models.

5. Discussion

5.1 Model Interpretability

Across all modelling approaches tested in our analysis, XGBoost consistently achieved the strongest predictive performances across both required predictions. However, the interpretability of an XGBoost model is lower compared to a linear model for instance. To address this, SHAP values were used in order to gain an insight into what features were most important.

The SHAP summary plot showed that the key features included lunch status, experience, gender, and class type. These findings align with our overall hypothesis that apart from teaching quality, socioeconomic status (captured by lunch and other such predictors), was a driver of academic outcomes. However, the magnitude and direction of these effects can vary from student to student, highlighting important heterogeneity that linear models struggle to capture.

In contrast, the linear models, be it OLS, Ridge, LASSO or Elastic Net provided clearer coefficient-based interpretation but were considerably worse at predicting scores accurately. Their reduced performance suggests that relationships between demographic factors and test scores are non-linear and involve interactions beyond what our modeling attempts to cover such that tree-based models capture them more effectively.

Overall, XGBoost offers a stronger predictive tool but requires post-hoc interpretability (e.g., SHAP), whereas linear models offer clearer interpretability but at the cost of predictive power.

5.2 Caveats

While the results offer meaningful insights, several limitations should be acknowledged:

1. **Missing Data Handling:** Our approach relies on dropping some rows and imputing/selectively filling missing values. These decisions may introduce bias if the missingness mechanism is not completely random.
2. **Feature Engineering Choices:** High-cardinality identifiers (school_id, district_id) and birth information were removed for valid modelling reasons, but doing so may obscure school-level structural effects that were not captured by the remaining variables.
3. **Limited Observables :** The dataset lacks many potentially important predictors of student performance (e.g., parental education, home environment, tutoring access). This restricts the maximum achievable model performance.
4. **Model Generalisability :** XGBoost may partially overfit to the training distribution despite cross-validation. Performance on truly unseen populations (different schools/districts) may vary.
5. **Interpretability Limits:** SHAP provides local and global explanations, but these are still approximations of the model's internal structure and should not be interpreted as causal effects.
6. **Multicollinearity in Linear Models:** The OLS and regularized models exhibited evidence of multicollinearity, which can distort coefficient estimates and weaken inference.

7. Limitations of Grid Search: Grid search is inherently constrained by the predefined parameter ranges. The algorithm only explores the specific values provided in the search space; therefore, if the selected ranges are too narrow or misaligned with the true optimal values, the resulting parameter combination may not yield the best possible model performance.

Together, these caveats highlight that while the models provide useful predictive and interpretive insights, they should be used cautiously and not treated as causal estimators.

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